DATA

Parameters in notable artificial intelligence systems

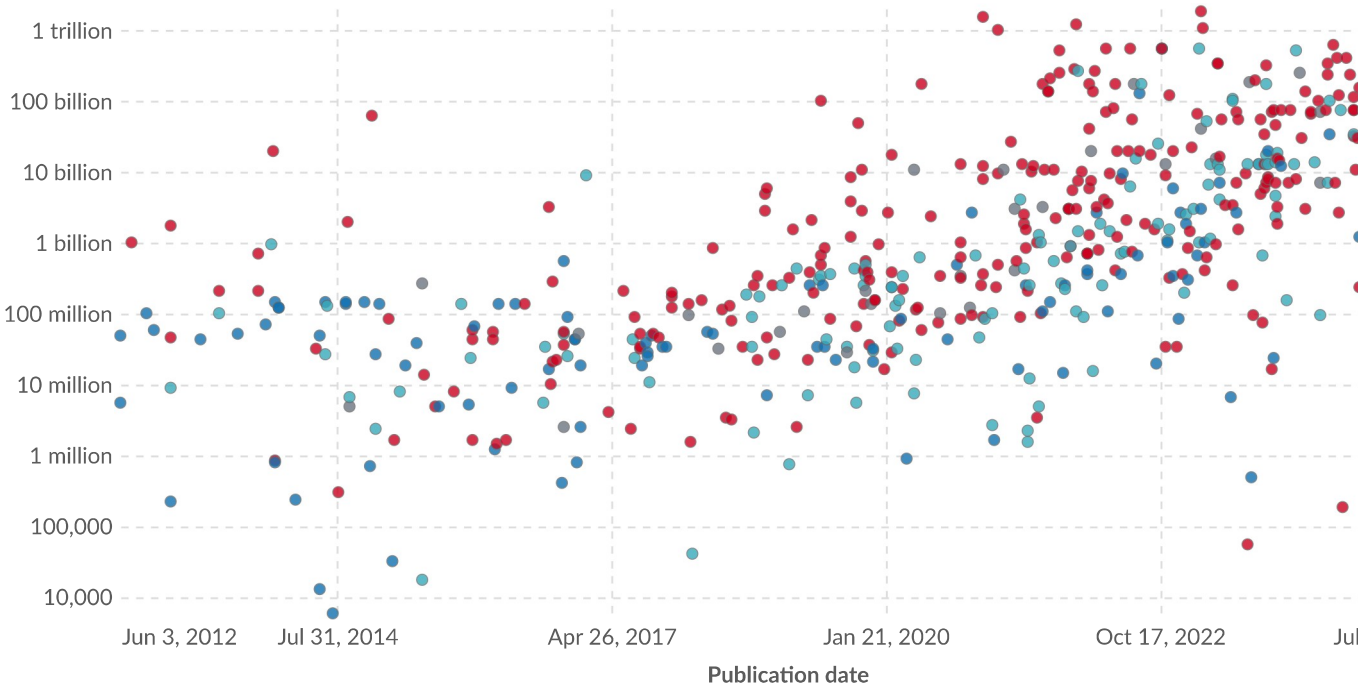
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Parameters in notable artificial intelligence systems

Parameters are variables in an AI system whose values are adjusted during training to establish how input data g
transformed into the desired output; for example, the connection weights in an artificial neural network.

Table Chart

Number of parameters (plotted on a logarithmic axis)



Play time-lapse Jul 2, 1950

Data source: Epoch AI (2025) – Learn more about this data OurWorldin

Note: Parameters are estimated based on published results in the AI literature and come with some uncertainty. The authors expect the est
be correct within a factor of 10.

What you should know about this indicator

- Parameters are internal variables that machine learning models adjust during their training process to improve their ability to make accurate predictions. They act as the model's "knobs" that are fine-tuned based on the provided data. In deep learning, a subset of artificial intelligence (AI), parameters primarily consist of the weights assigned to the connections between the small processing units called neurons. Picture a vast network of interconnected neurons where the strength of each connection represents a parameter.

PARAMETERS IN NOTABLE ARTIFICIAL INTELLIGENCE SYSTEMS

Total number of learnable variables or weights that the model contains. Parameters are adjusted during the

connection represents a parameter.

- The total number of parameters in a model is influenced by various factors. The model's structure and the number of “layers” of neurons play a significant role. Generally, more complex models with additional layers tend to have a higher number of parameters. Special components of specific deep learning architectures can further contribute to the overall parameter count.
- Understanding the number of parameters in a model is crucial to design effective models. More parameters can help the model understand complex data patterns, potentially leading to higher accuracy. However, there's a fine balance to strike. If a model has too many parameters, it risks memorizing the specific examples in its training data rather than learning their underlying patterns. Consequently, it may perform poorly when presented with new, unseen data. Achieving the right balance of parameters is a critical consideration in model development.
- In recent times, the AI community has witnessed the emergence of what are often referred to as "giant models." These models boast an astounding number of parameters, reaching into the billions or even trillions. While these huge models have achieved remarkable performance, they have a significant computational cost. Effectively managing and training such large-scale models has become a prominent and active area of research and discussion within the AI field.

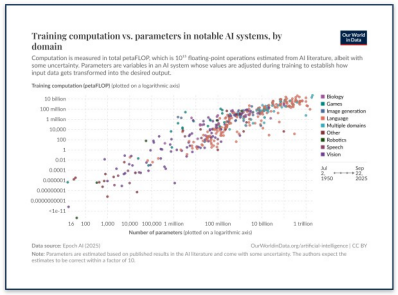
training process to optimize the model's performance.

Source
Epoch AI (2025) – with major processing by Our World in Data

Last updated
March 12, 2025

Next expected update
January 2026

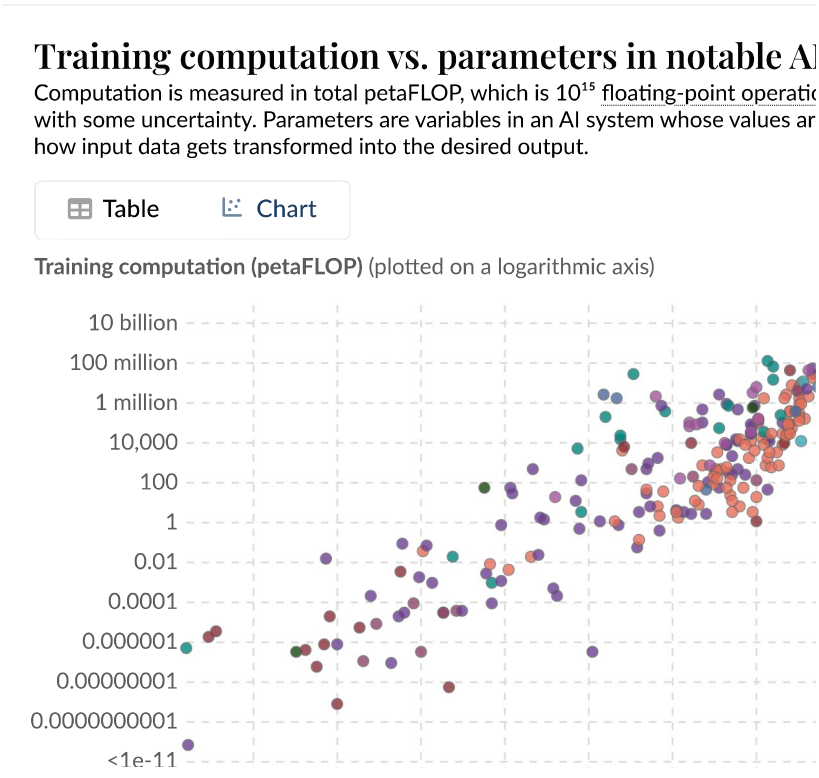
Explore charts that include this data



Training computation vs. parameters in notable AI systems, by domain



Training co
in notable A
affiliation





Jul 2, 1950

Data source: Epoch AI (2025) – Learn more about this data

Note: Parameters are estimated based on published results in the AI literature and come
expect the estimates to be correct within a factor of 10.

Chart 1 of 3

Sources and processing

This data is based on the following sources

Epoch AI – Parameter, Compute and Data Trends in Machine Learning +

Retrieved on
October 9, 2025

Retrieved from
https://epoch.ai/mlinputs/visualization

Citation

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Read about our data pipeline →

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HOW TO CITE THIS DATA

In-line citation

If you have limited space (e.g. in data visualizations), you can use this abbreviated in-line citation:

Epoch AI (2025) – with major processing by Our World in Data

Full citation

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